

STORA SPANSBERGET, Sweden

WMO: 024680

Lat: 60.382N

Lon: 15.137E

Elev: 1617

StdP: 13.86

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	6.8	-2.2	5.2	3.6	2.2	6.5	9.4	18.6	25.7	16.5	26.4	3.5	20	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	76.1	61.1	72.7	59.8	69.5	58.3	64.0	71.5	62.0	69.0	60.2	66.4	5.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	86.3	65.4	59.4	80.4	64.3	57.7	75.4	62.4	30.1	71.8	28.6	68.8	27.2	66.4	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	14.1	12.4	DB	-0.6	80.4	5.3	2.5	-4.4	82.2	-7.5	83.7	-10.4	85.0	-14.2	86.9
			WB	-1.1	66.1	5.1	2.8	-4.7	68.1	-7.7	69.7	-10.5	71.3	-14.2	73.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	22.6	23.0	28.6	38.1	47.7	55.1	59.7	56.8	48.8	38.8	30.8	25.6	(6)
	DBStd	14.78	7.72	7.92	7.79	6.86	7.33	6.37	5.95	5.14	5.30	6.33	7.00	7.99	(7)
	HDD50	4560	850	756	663	363	133	20	2	6	82	351	577	757	(8)
	HDD65	9261	1315	1176	1128	808	537	303	188	258	485	814	1027	1222	(9)
	CDD50	809	0	0	0	5	62	174	301	218	47	2	0	0	(10)
	CDD65	35	0	0	0	0	0	7	23	5	0	0	0	0	(11)
	CDH74	189	0	0	0	0	4	50	118	17	0	0	0	0	(12)
	CDH80	15	0	0	0	0	0	1	12	2	0	0	0	0	(13)
(14) Wind	WSAvg	6.2	5.8	5.9	7.0	6.6	6.3	5.9	5.6	5.4	6.1	6.9	6.9	6.0	(14)
(15) Precipitation	PrecAvg	26.6	1.7	1.3	1.3	1.6	2.0	2.9	3.4	3.3	2.3	2.5	2.2	2.0	(15)
	PrecMax	34.4	3.6	2.4	2.6	4.1	5.2	5.7	9.1	6.3	5.0	6.4	5.8	3.7	(16)
	PrecMin	21.5	0.3	0.2	0.1	0.2	0.4	0.6	0.9	0.6	0.4	0.7	0.8	0.3	(17)
	PrecStd	3.4	0.8	0.6	0.7	1.0	1.1	1.2	1.8	1.4	1.2	1.3	1.2	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	40.5	51.7	62.8	73.8	78.7	81.7	76.1	66.0	56.3	47.9	43.6	(19)
		MCWB	37.2	35.4	41.6	49.4	55.8	60.3	64.1	61.8	57.2	52.6	46.7	43.2	(20)
	2%	DB	36.4	37.9	45.9	57.6	68.9	75.0	77.1	72.0	62.1	51.9	45.0	40.1	(21)
		MCWB	35.2	35.0	38.4	46.2	53.9	59.0	62.8	60.8	55.1	50.1	43.7	39.2	(22)
	5%	DB	34.4	35.5	42.1	53.0	64.6	70.9	73.7	68.7	59.1	49.7	42.4	37.8	(23)
		MCWB	33.6	33.7	36.4	43.5	52.6	57.3	61.4	59.4	53.8	48.3	41.7	36.9	(24)
	10%	DB	32.6	33.4	38.8	48.8	59.9	66.7	70.3	65.3	56.2	47.6	40.2	35.4	(25)
		MCWB	32.0	32.1	34.4	40.8	49.8	55.0	59.9	58.1	52.4	46.4	39.8	34.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	38.3	42.2	50.7	59.1	63.7	67.2	65.4	59.3	53.5	46.5	43.4	(27)
		MCDB	38.2	39.3	49.7	60.4	67.2	73.1	75.0	71.1	63.1	55.2	46.6	43.4	(28)
	2%	WB	35.4	35.6	39.3	47.4	56.6	60.7	64.9	62.8	56.6	50.8	43.3	39.6	(29)
		MCDB	36.0	36.6	44.2	55.9	65.8	69.6	73.0	68.2	60.3	51.4	43.9	39.9	(30)
	5%	WB	33.8	33.9	37.2	44.5	53.8	58.5	63.2	60.7	54.9	48.7	41.3	37.0	(31)
		MCDB	34.4	35.2	41.2	51.3	61.7	67.6	71.1	65.6	57.9	49.3	41.8	37.5	(32)
	10%	WB	32.1	32.1	35.2	41.8	51.4	56.4	61.3	59.1	53.1	46.8	39.2	34.4	(33)
		MCDB	32.5	33.3	38.0	47.2	59.3	64.7	68.5	64.0	55.5	47.6	39.7	35.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.6	10.1	13.1	15.5	15.9	14.8	12.8	10.7	7.9	6.1	6.6	(35)
		MCDBR	8.0	8.3	13.8	18.6	21.4	21.3	19.4	17.0	14.0	9.6	8.4	8.5	(36)
	5% WB	MCWBR	8.0	6.6	8.7	10.5	11.5	10.4	9.1	8.6	8.3	7.7	8.7	8.5	(37)
		MCWBR	7.6	7.7	12.5	17.6	19.0	19.5	17.7	14.2	12.4	8.8	8.5	8.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.238	0.282	0.306	0.340	0.340	0.334	0.353	0.346	0.315	0.302	0.259	0.212	(40)
		taud	2.214	2.336	2.405	2.414	2.461	2.499	2.457	2.470	2.548	2.497	2.280	2.096	(41)
	Ebn at Noon	Ebn at Noon	175	232	264	273	281	285	276	269	260	224	164	131	(42)
		Edh at Noon	12	20	25	30	31	30	31	29	22	17	11	8	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	88	295	780	1230	1510	1715	1566	1218	802	363	111	55	(44)
	RadStd	RadStd	16	55	92	143	153	115	191	133	111	66	30	7	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page